Chapter 7

The Singular Value Decomposition

Homework list

Strang

Sec 7.1: 1, 6, 7

Sec 7.2: 1, 2, 16

Sec 7.3: 1, 4 (Hint: The SVD of A shows the dominant direction in the scatter point. The best fit line through the origin is the line defined by the singular vector of the largest singular value).

Sec 7.4: 5, 6, 7, 10, 18

Section 7.1

Image Processing by Linear Algebra

Data matrices.

- **Matrices also arise in many situations in which we tend not to think of an associate linear system; they are simply rectangular arrays of numbers.**
- **Grading data from class (made up numbers):**

	Homework	Quiz	Exam
Student 1	78%	84%	67%
Student 2	85%	76%	72%
Student 3	100%	100%	93%
Student 4	100%	40%	31%
Student 5	91%	95%	86%
Student 6	92%	87%	62%

$$\rightarrow A = \begin{bmatrix} 0.78 & 0.84 & 0.67 \\ 0.85 & 0.76 & 0.72 \\ 1.00 & 1.00 & 0.93 \\ 1.00 & 0.40 & 0.31 \\ 0.91 & 0.95 & 0.86 \\ 0.92 & 0.87 & 0.62 \end{bmatrix}$$

Big Data in Biology

- **Many "high-throughput" experiments in biology generate big data matrices:**
- **Example: Expression levels of genes in different cell types**

	Cell Type 1	Cell Type 2	⋯	Cell Type 200
Gene 1	12501	3281		2983
Gene 2	5707	6347		6532
Gene 3	102	8002		7692
Gene 4	11597	12432		11832
⋮				
Gene 30,000	5	7		11

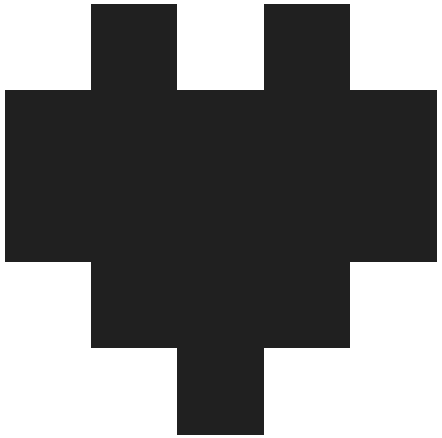
Big Data in Finance and Economics

- **Financial and economic time series data naturally lends itself to representation in large matrices:**
- **Example: National GDP growth over time.**

	US	Canada	Mexico	⋯
2020 Q2	-31.4%	-13.0%	-18.7%	
2020 Q1	-5.0%	-0.9%	-2.1%	
2019 Q4	+2.4%	+1.5%	-0.8%	
2019 Q3	+2.6%	+1.6%	-0.5%	
2019 Q2	+1.5%	+2.0%	0.0%	
⋮				

Big Data in Image Processing

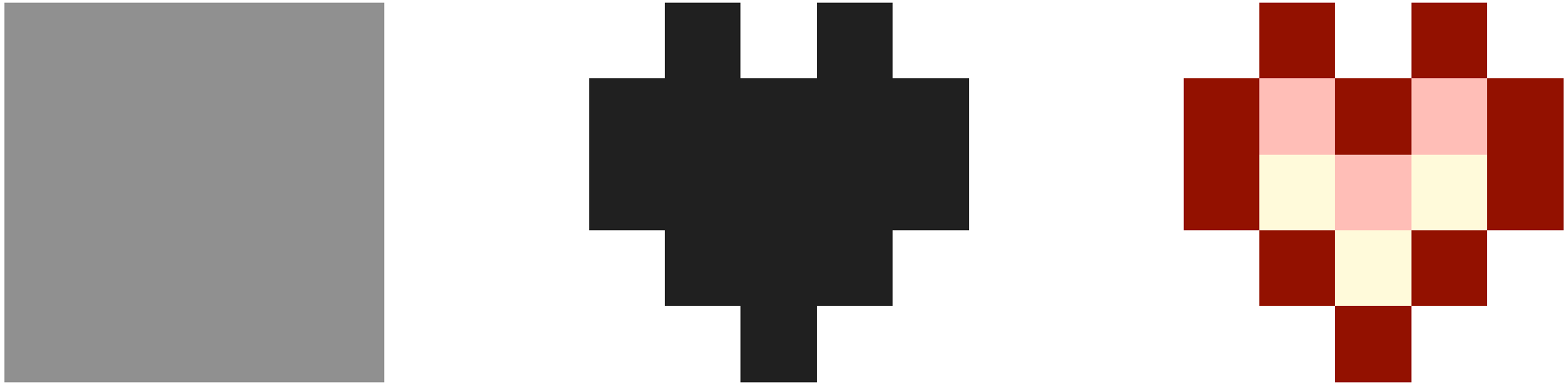
- **Digital images are rectangular arrays of numbers representing color and/or intensity.**
- **A pixel is a "picture element" → one element in the array.**
- **A 12 megapixel camera records images as 4092x3072 arrays!**



$$\rightarrow A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

Data, information and knowledge

- **Example: Images of the exactly same size can contain contain very different amounts of information.** *An image is a large matrix of grayscale values, one for each pixel and color. When nearby pixels are correlated (not random) the image can be compressed.*



Extreme example

The easiest images to compress are all black or all white or all a constant grayscale g. The matrix A has the same number g in every entry: $a_{ij} = g$. When $g = 1$ and $m = n = 6$, here is an extreme example of the central SVD dogma of image processing:

Example 1 **Don't send** $A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$ **Send this** $A = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$

36 numbers become 12 numbers. With 300 by 300 pixels, 90,000 numbers become 600. And if we define the all-ones vector $\boldsymbol{x}$ in advance, we only have to send **one number**. That number would be the constant grayscale g that multiplies $\boldsymbol{x}\boldsymbol{x}^{\mathrm{T}}$ to produce the matrix.

Example: Don’t send and send

Example 2
“ace flag”
French flag $\boldsymbol{A}$
Italian flag $\boldsymbol{A}$
German flag $\boldsymbol{A}^{\mathbf{T}}$

Don’t send $A = \begin{bmatrix} a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \\ a & a & c & c & e & e \end{bmatrix}$ **Send** $A = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} a & a & c & c & e & e \end{bmatrix}$

Data, information and knowledge

- **One of the central problems in data science is how to translate from data to knowledge and understanding.**
- **A big question: how much information is actually in a data set?**
  - **What are the "essential features"?**
  - **What are the subtle details?**
  - **What is random noise (error)?**

The ideas are essential to building intuitive understanding, to recognizing similarities between sets of data and in efficient storage.

Rank theorem: Matrices as sums of rank-1 matrices.

Every matrix of rank r can be written as a sum of r rank-1 matrices:

Examples:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & -1 & -1 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 5 \\ 2 & 3 & 5 \\ 3 & 2 & 5 \\ 4 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 3 & 0 & 3 \\ 4 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 4 & 4 \\ 0 & 3 & 3 \\ 0 & 2 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

Matrices as sums of rank-1 matrices.

This sum is not unique:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Review: Rank and Nullity of a matrix A

Definition

Let A be an m by n matrix.

1. The **rank** of A, denoted rank(A) is the number of linearly independent columns of A (equivalently the number of linearly independent rows).
2. The **nullity** of A denoted nullity(A) is given by n−rank(A). It gives the dimension of the null space of A, i.e. the size of the largest linearly independent set of vectors $v \in R^n$ such that Av =0.

- Note : Rank (A) $\leq \min(m, n)$.

Example 3 from Strang's book

Example 3
Embedded square

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \text{ is equal to } A = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}$$

Further observation: Rank-1 matrices are "outer products".

- The inner (dot) product is $\vec{u}^T\vec{v}$, a scalar; the outer product is $\vec{u}\vec{v}^T$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 3 & 0 & 3 \\ 4 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$$

Matrices as sums of outer products

- **Every matrix of rank r can be written as a sum of r outer products:**

$$rank(A) = r, A \in \mathbb{R}^{m\times n} \rightarrow A = \sum_{i=1}^{r} \vec{x}_i \vec{y}_i^T, \vec{x}_i \in \mathbb{R}^m, \vec{y}_i \in \mathbb{R}^n$$

- **Every vector can be written as a linear combination of orthonormal basis vectors, $\mathbb{R}^m = span(\{\vec{u}_i\})$ and $\mathbb{R}^n = span(\{\vec{v}_i\})$:**

$$\vec{x}_i \in \mathbb{R}^m \rightarrow \vec{x}_i \in \sum_{j=1}^{m} a_i \vec{u}_i, \qquad \vec{y}_i \in \mathbb{R}^n \rightarrow \vec{y}_i \in \sum_{j=1}^{n} b_i \vec{v}_i$$

- **Every matrix of rank r can be written as a linear combination of r outer products of orthonormal vectors with positive coefficients.**

$$A = \sum_{i=1}^{r} \sigma_i \vec{u}_i \vec{v}_i^T, \sigma_i > 0$$

!!!!!This is the key of the singular value decomposition!!!!!

U and V?

By rank theorem, any m by n matrix A with rank r can be written as the weighted summation r rank-1 matrices.

$$A = \sum_{i=1}^{r} \sigma_i \vec{u}_i \vec{v}_i^T \rightarrow A = U\Sigma V^T$$

where $\mathbf{u_i}$ and **$\mathbf{v_i}$** are the ith column of matrix **U** and **V** respectively. This *m* by *n* matrix $\mathbf{u_i}\,\mathbf{v^T_i}$ is the product of a column vector $\mathbf{u_i}$ and the transpose of column vector $\mathbf{v_i}$. It has rank 1. Thus, **A** is a weighted summation of *r* rank-1 matrices.

Key question: How could we find U and V? Not all orthonormal bases work here. Only special ones work.

Construction of singular vectors for SVD

Only special orthonormal bases work to give $A = U\Sigma V^T$.

Step1: Amazing construction of u's and v's.

- **The product of a matrix and its transpose is always symmetric:**

$$S = A^T A, \qquad S^T = S$$

- **Symmetric matrices can always be diagonalized with orthonormal eigenvectors:**

$$S = A^T A = Q\Lambda Q^T$$

$$A^T A = (U\Sigma V^T)^T U\Sigma V^T = V\Sigma^T\Sigma V^T \rightarrow Q = V, \Sigma^T\Sigma = \Lambda$$

- **The singular vectors $\{\vec{v}_i\}$ are the eigenvectors of $A^T A$.**
- **The eigenvalues of $A^T A$ are the singular values squared, $\lambda_i = \sigma^2$.**

Step 2

Step2: The product of a matrix and its transpose is always symmetric:

$$S = \mathrm{A}A^T, \qquad S^T = S$$

- **Symmetric matrices can always be diagonalized with orthonormal eigenvectors:**

$$S = \mathrm{A}A^T = Q\Lambda Q^T$$

- **Similarly, the singular vectors u_i are the eigenvectors of $A^T A$.**

- **The eigenvalues of $A^T A$ are the singular values squared, $\lambda_i = \sigma^2$.**

Definition of singular values for Σ

Definition

The singular values $\sigma_1, \sigma_2, \ldots, \sigma_n$ of an $m \times n$ matrix A are the square roots of the eigenvalues of $A^T A$.

Note that this means that the rank of A is equal to the number of non-zero singular values.

Important Facts/Theorems

1. The matrices $A^T A$ and AA^T are symmetric, square and semidefinite.

2. nullity(A)=nullity($A^T A$)
3. rank(A)=rank($A^T A$)
4. The eigenvalues of $A^T A$ and AA^T are real and nonnegative
5. The nonzero eigenvalues of $A^T A$ and AA^T coincide.

Proof

1. It is easy to check the transpose of $A^T A$ and AA^T remain the same. $x^T A^T A x = (Ax)^T (Ax) = ||Ax||^2 \geq 0 \;\; for\ any\ x \neq 0$. Then $A^T A$ and AA^T are symmetric, square and positive semidefinite.

2. Let $v \in \mathbb{R}^n$ be such that $Av = 0$. Then it is clear that $A^T Av = 0$. Now, if $v \in \mathbb{R}^n$ such that $A^T Av = 0$ we have

$$0 = v^T A^T Av = \|Av\|_2^2 = 0.$$

This then implies that $Av = 0$. Since the two matrices share the same null vectors (or nullspace) the nullity must agree.

3. This follows directly from the definition of rank because the two matrices have the same number of columns: $n - \text{nullity}(A) = n - \text{nullity}(A^T A)$.

Continued

4. This is the conclusion from the spectral theorem since $A^T A$ and AA^T are symmetric matrices.

5. Assume for $v \neq 0$, $\lambda \neq 0$, $A^T Av = \lambda v$. Then $AA^T Av = \lambda Av$. Let $w = Av$ and we have $AA^T w = \lambda w$. So, if $w \neq 0$ then λ is an eigenvalue of AA^T and w cannob be zero as $A^T Av \neq 0 \Rightarrow Av \neq 0$. Now assume $AA^T w = \lambda w$ for $\lambda \neq 0$ and $w \neq 0$. Then $A^T AA^T w = \lambda A^T w$. Let $v = A^T w$ and we have $A^T Av = \lambda v$ and we can conclude that λ is an eigenvalue if $v \neq 0$. Again, if $v = 0$ then $A^T Aw = 0$ which contradicts $\lambda, w \neq 0$. So, λ is an eigenvalue of $A^T A$.

Formal Definition: The Singular Value Decomposition

Define two square matrices whose columns are orthonormal bases for $\mathbb{R}^m$ and $\mathbb{R}^n$:

$$U = [\vec{u}_1 \quad \vec{u}_2 \quad \cdots \quad \vec{u}_m], V = [\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n]$$

Define a diagonal matrix of the same size as A:

$$\Sigma = diag([\sigma_1, \sigma_2, \cdots, \sigma_r])$$

- **Then:**

$$A = \sum_{i=1}^{r} \sigma_i \vec{u}_i \vec{v}_i^T \rightarrow A = U\Sigma V^T$$

- **Only very special orthonormal bases work!**

Usefulness of SVD

$$A = U\Sigma V^T$$

- **A matrix decomposition that is particularly useful in dealing with data matrices.**
  - **Can be used to find "reduced rank" approximations of matrices.**
  - **Can be used to identify the "principal components" of a data set.**
- **Also applicable to solving linear systems of the types we have already seen:**
  - **Can also be used to compute a pseudo-inverse for matrices that are neither full column nor full row rank.**
  - **Can be used to minimize the error when computing the inverse of "near singular" matrices.**
  - **The SVD separates any matrix A into rank one pieces uv^T = (column)(row). The Singular Value Decomposition (SVD) separates any matrix into simple pieces.**

Connections between SVD and EVD

Rationale: Generally, we cannot directly diagonalize an mxn matrix A using $X^{-1}AX$ where X are the eigenvectors of A because (1) the eigenvectors are usually not orthogonal, (2) there may not be enough eigenvectors, and (3) $Ax = \lambda x$ requires A to be a square matrix. Due to the limitation of the eigenvalue decomposition, we need to find a more general decomposition that works for any matrix A.

Connection: The singular value decomposition for A is the eigenvalue decomposition for $A^T A$ and AA^T .

Key ideas:

1. A has two sets of singular vectors (eigenvectors of $A^T A$ and AA^T).
2. One set of positive singular values (because $A^T A$ has the same positive eigenvalues as AA^T).
3. A can be rectangular, but $A^T A$ and AA^T are square, symmetric, and positive semidefinite.
4. We can construct an orthonormal set of eigenvectors.

Connections: Singular vectors and eigenvectors

Symmetric S	$S = Q\Lambda Q^{\mathrm{T}} \quad = \lambda_1 \boldsymbol{q}_1 \boldsymbol{q}_1^{\mathrm{T}} + \lambda_2 \boldsymbol{q}_2 \boldsymbol{q}_2^{\mathrm{T}} + \cdots + \lambda_r \boldsymbol{q}_r \boldsymbol{q}_r^{\mathrm{T}}$
Any matrix A	$A = U\Sigma V^{\mathrm{T}} \quad = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathrm{T}} + \sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^{\mathrm{T}} + \cdots + \sigma_r \boldsymbol{u}_r \boldsymbol{v}_r^{\mathrm{T}}$

The singular values σ_i of A are the square roots of the eigenvalues λ_i of $S = A^{\mathrm{T}}A$. This connects the SVD to a *symmetric eigenvalue problem* (good). But in the end we don't want to multiply A^{T} times A (squaring is time-consuming: not good).

Real applications of Singular value decomposition.

- In machine learning, SVD is often used for dimensionality reduction, data compression, and denoising. The goal is to reduce the number of dimensions in the data set while preserving as much information as possible. By reducing the dimensionality of the data, SVD helps to mitigate the curse of dimensionality, which is a common problem in machine learning, where the performance of algorithms decreases as the number of features in the data increases.
- Singular-value decomposition (SVD) allows an exact representation of any matrix and it is easy to eliminate the less important data in the matrix to produce a low-dimensional approximation. This is meaningful in such applications as image compression and recommendation system.

Example

In Example 3 (the embedded square), here are the symmetric matrices AA^{T} and $A^{\mathrm{T}}A$:

$$AA^{\mathrm{T}} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{2} \end{bmatrix} \qquad A^{\mathrm{T}}A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{2} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} \end{bmatrix}.$$

Their determinants are 1, so $\lambda_1\lambda_2 = 1$. Their traces (diagonal sums) are 3:

$$\det \begin{bmatrix} 1-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = \lambda^2 - 3\lambda + 1 = 0 \quad \text{gives} \quad \lambda_1 = \frac{3+\sqrt{5}}{2} \quad \text{and} \quad \lambda_2 = \frac{3-\sqrt{5}}{2}.$$

The square roots of λ_1 and λ_2 are $\sigma_1 = \dfrac{\sqrt{5}+1}{2}$ and $\sigma_2 = \dfrac{\sqrt{5}-1}{2}$ with $\sigma_1\,\sigma_2 = 1$.

The nearest rank 1 matrix to A will be $\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^{\mathrm{T}}$. The error is only $\sigma_2 \approx 0.6 =$ best possible.

The orthonormal eigenvectors of AA^{T} and $A^{\mathrm{T}}A$ are

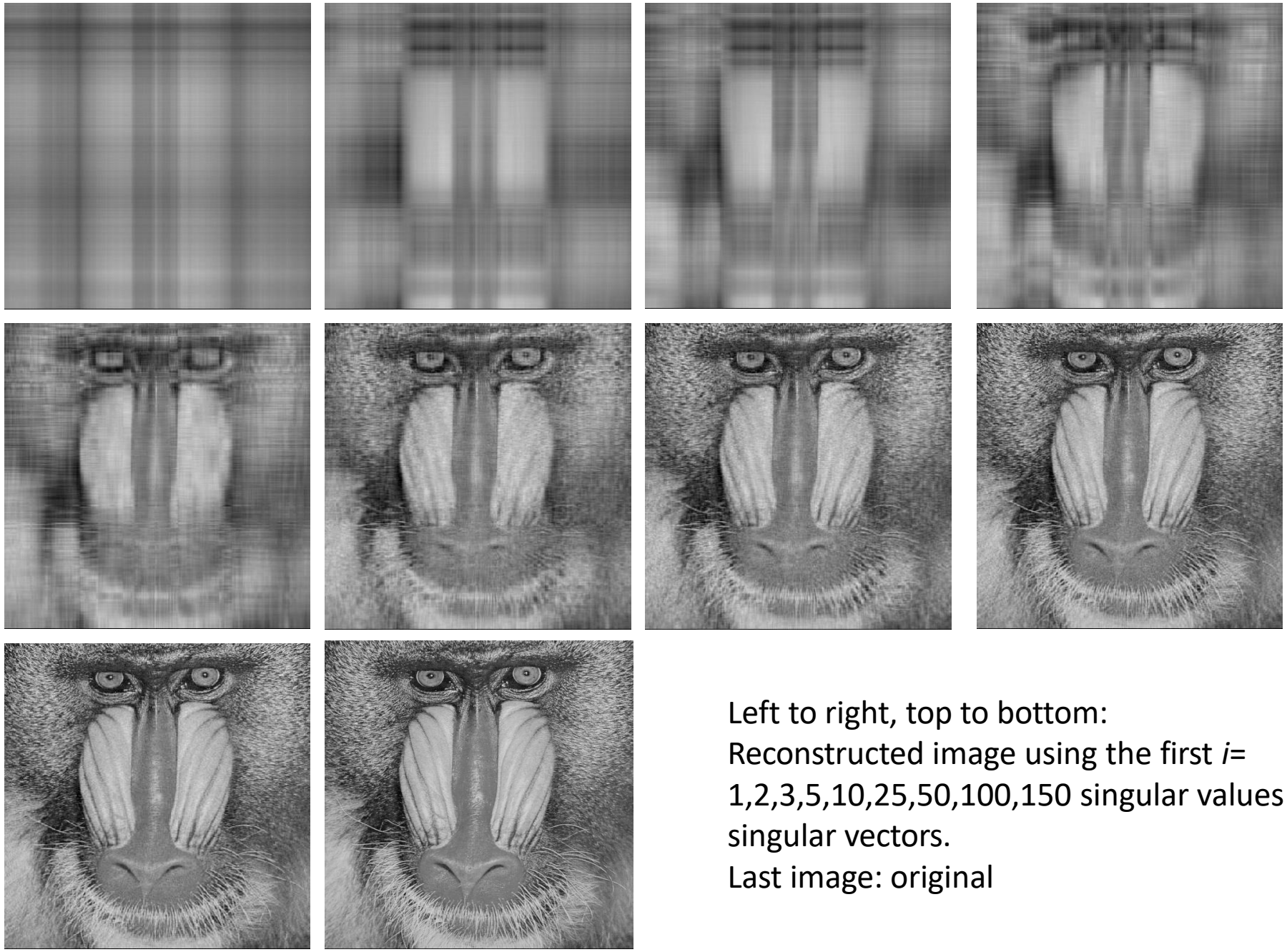
$$\boldsymbol{u}_1 = \begin{bmatrix} 1 \\ \sigma_1 \end{bmatrix} \quad \boldsymbol{u}_2 = \begin{bmatrix} \sigma_1 \\ -1 \end{bmatrix} \quad \boldsymbol{v}_1 = \begin{bmatrix} \sigma_1 \\ 1 \end{bmatrix} \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ -\sigma_1 \end{bmatrix} \quad \text{all divided by } \sqrt{1+\sigma_1^2}. \qquad (2)$$

Application: SVD of Natural Images

- An image is a 2D array – each entry contains a grayscale value. The image can be treated as a matrix.
- It has been observed that for many image matrices, the singular values undergo rapid decay (note: they are always non-negative).
- An image can be approximated with the k largest singular values and their corresponding singular vectors:

SVD: Use in Image Compression

- Instead of storing *mn* intensity values, we store *(n+m+1)r (any m by n matrix A with rank r can be written as summation of r outer products of m-dim and n-dim vectors, so (n+m)r values. There are additional r singular values)* intensity values where *r* is the number of stored singular values (or singular vectors). The remaining *m-r* singular values (and hence their singular vectors) are effectively set to 0.
- This is called as storing a low-rank (rank *r*) approximation for an image.



Left to right, top to bottom:
Reconstructed image using the first *i*= 1,2,3,5,10,25,50,100,150 singular values and singular vectors.
Last image: original

Other applications of SVD

- **Face recognition** –the popular eigen faces algorithm can be implemented using SVD too!
- **Point matching:** Consider two sets of points, such that one point set is obtained by an unknown rotation of the other. Determine the rotation!
- **Structure from motion:** Given a sequence of images of a object undergoing rotational motion, determine the 3D shape of the object as well as the 3D rotation at every time instant!

Section 7.2

Bases and Matrices in Singular Value Decomposition (SVD)

Dimensional Analysis of SVD

For any *m* x *n* matrix **A**, the following decomposition **always** exists:

$$A = U\Sigma V^T, A \in R^{m\times n},\ \ U^T U = U U^T = I_m, U \in R^{m\times m},$$

$$V^T V = V V^T = I_n, V \in R^{n\times n},\ \ \Sigma \in R^{m\times n}$$

- Columns of **U** are the eigenvectors of $\mathbf{AA}^T$ (called the left singular vectors).
- Columns of **V** are the eigenvectors of $\mathbf{A}^T\mathbf{A}$ (called the right singular vectors).
- The non - zero singular values are the positive square roots of non - zero eigenvalues of $\mathbf{AA}^T$ or $\mathbf{A}^T\mathbf{A}$.

Facts of Singular Value Decomposition

- For any *m* x *n* real matrix **A**, the SVD consists of matrices **U**, Σ,**V** which are always real – this is unlike eigenvectors and eigenvalues of **A** which may be complex even if **A** is real.

- The singular values are always non-negative, even though the eigenvalues may be negative.

- While writing the SVD, the following convention is assumed, and the left and right singular vectors are also arranged accordingly:

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_{m-1} \geq \sigma_m$$

SVD: the right bases for the four subspaces

(using vectors) The $\boldsymbol{u}$'s and $\boldsymbol{v}$'s give bases for the four fundamental subspaces:

$\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r$	is an orthonormal basis for the **column space**
$\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$	is an orthonormal basis for the **left nullspace** $\boldsymbol{N}(A^{\mathrm{T}})$
$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r$	is an orthonormal basis for the **row space**
$\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n$	is an orthonormal basis for the **nullspace** $\boldsymbol{N}(A)$.

More than just orthogonality, these basis vectors diagonalize the matrix A:

"A is diagonalized" $\quad A\boldsymbol{v}_1 = \sigma_1 \boldsymbol{u}_1 \quad A\boldsymbol{v}_2 = \sigma_2 \boldsymbol{u}_2 \quad \ldots \quad A\boldsymbol{v}_r = \sigma_r \boldsymbol{u}_r \qquad (1)$

Those **singular values σ_1 to σ_r** will be positive numbers: *σ_i is the length of $A\boldsymbol{v}_i$*. The σ's go into a diagonal matrix that is otherwise zero. That matrix is Σ.

An important property: $Av_i = \sigma u_i$.

The $\boldsymbol{u}$'s from the SVD are called **left singular vectors** (unit eigenvectors of AA^{T}). The $\boldsymbol{v}$'s are **right singular vectors** (unit eigenvectors of $A^{\mathrm{T}}A$). The σ's are **singular values**, square roots of the equal eigenvalues of AA^{T} and $A^{\mathrm{T}}A$:

Choices from the SVD $\quad AA^{\mathrm{T}}\boldsymbol{u}_i = \sigma_i^2 \boldsymbol{u}_i \quad A^{\mathrm{T}}A\boldsymbol{v}_i = \sigma_i^2 \boldsymbol{v}_i \quad A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i \qquad (1)$

$$A = \begin{bmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1^{\mathrm{T}} \\ \boldsymbol{v}_2^{\mathrm{T}} \end{bmatrix} \text{ or more simply } A \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 \end{bmatrix} = \begin{bmatrix} \sigma_1 \boldsymbol{u}_1 & \sigma_2 \boldsymbol{u}_2 \end{bmatrix} \qquad (3)$$

(using matrices) Since the $\boldsymbol{u}$'s are orthonormal, the matrix U_r with those r columns has $U_r^{\mathrm{T}} U_r = I$. Since the $\boldsymbol{v}$'s are orthonormal, the matrix V_r has $V_r^{\mathrm{T}} V_r = I$. Then the equations $A\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i$ tell us column by column that $\boldsymbol{AV_r} = \boldsymbol{U_r \Sigma_r}$:

$$\begin{array}{c} (m \text{ by } n)(n \text{ by } r) \\ \boldsymbol{AV_r} = \boldsymbol{U_r \Sigma_r} \\ (m \text{ by } r)(r \text{ by } r) \end{array} \qquad A \begin{bmatrix} \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_r \end{bmatrix} = \begin{bmatrix} \boldsymbol{u}_1 & \cdots & \boldsymbol{u}_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}. \tag{2}$$

$$\begin{array}{c} (m \text{ by } n)(n \text{ by } n) \\ \boldsymbol{AV} \text{ equals } \boldsymbol{U\Sigma} \\ (m \text{ by } m)(m \text{ by } n) \end{array} \qquad \boldsymbol{A} \begin{bmatrix} \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_r & \cdots & \boldsymbol{v}_n \end{bmatrix} = \begin{bmatrix} \boldsymbol{u}_1 & \cdots & \boldsymbol{u}_r & \cdots & \boldsymbol{u}_m \end{bmatrix} \begin{bmatrix} \boldsymbol{\sigma_1} & & \\ & \ddots & \\ & & \boldsymbol{\sigma_r} \\ & & \end{bmatrix} \tag{3}$$

The new Σ is m by n. It is just the r by r matrix in equation (2) with $m - r$ extra zero rows and $n - r$ new zero columns. The real change is in the shapes of U and V. Those are square matrices and $V^{-1} = V^{\mathrm{T}}$. So $AV = U\Sigma$ becomes $\boldsymbol{A} = \boldsymbol{U\Sigma V^{\mathrm{T}}}$. This is the ***Singular Value Decomposition***. I can multiply columns $\boldsymbol{u}_i \sigma_i$ from $U\Sigma$ by rows of V^{T}:

SVD $$\boldsymbol{A = U\Sigma V^{\mathrm{T}} = u_1 \sigma_1 v_1^{\mathrm{T}} + \cdots + u_r \sigma_r v_r^{\mathrm{T}}}. \tag{4}$$

Relationship between SVD and Four subspaces

- Only very special orthonormal bases work to give $A = U\Sigma V^T$.

$$A^T A = V\Sigma^T \Sigma V^T = Q\Lambda Q^T, Q = V, \Sigma^T \Sigma = \Lambda$$

- The singular vectors $\{\vec{v}_i\}$ are the eigenvectors of $A^T A$.
- The eigenvalues of $A^T A$ are the singular values squared, $\lambda_i = \sigma_i^2$.

Important!

- Singular vectors $\vec{v}_i$ with $\lambda_i = \sigma_i^2 = 0$ are in the null space of A:

$$A\vec{x} = \vec{0} \rightarrow A^T A\vec{x} = A^T\vec{0} = \vec{0} = 0\vec{x}$$

- Singular vectors $\vec{v}_i$ with $\lambda_i = \sigma_i^2 \neq 0$ are in the row space of A:

$$A^T A\vec{x} = \lambda\vec{x} \rightarrow A^T\vec{y} = \lambda\vec{x}, \vec{y} = A\vec{x} \rightarrow \lambda\vec{x} \in \boldsymbol{C}(A^T)$$

Relationship between SVD and Four subspaces

- **Singular vectors with $\sigma_i = 0$ are in the null space of A; singular vectors with $\sigma_i \neq 0$ are in the row space of A.**

$$AA^T = U\Sigma V^T (U\Sigma V^T)^T = U\Sigma\Sigma^T U^T = U\Lambda' U^T, \Sigma^T\Sigma = \Lambda'$$

- **The singular vectors $\{\vec{u}_i\}$ are the eigenvectors of AA^T with $\lambda_i = \sigma^2$.**
- **Singular vectors $\vec{u}_i$ with $\sigma^2 = 0$ are in the left null space of A.**
- **Singular vectors $\vec{u}_i$ with $\sigma^2 \neq 0$ are in the column space of A.**

Relationship between SVD and Four subspaces

- **Singular vectors $\vec{v}_i$ with $\sigma_i = 0$ are in the null space of A; singular vectors $\vec{v}_i$ with $\sigma_i \neq 0$ are in the row space of A:**

$$V = [V_R \quad V_N], V_R = [\vec{v}_1 \quad \cdots \quad \vec{v}_r], V_N = [\vec{v}_{r+1} \quad \cdots \quad \vec{v}_n]$$

- **Singular vectors $\vec{u}_i$ with $\sigma_i = 0$ are in the left null space of A; singular vectors $\vec{u}_i$ with $\sigma_i \neq 0$ are in the column space of A.**

$$U = [U_C \quad U_{LN}], U_C = [\vec{u}_1 \quad \cdots \quad \vec{u}_r], U_{LN} = [\vec{u}_{r+1} \quad \cdots \quad \vec{u}_m]$$

$$\Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

Relationship between SVD and Four subspaces

- **Only very special orthonormal bases work to give $A = U\Sigma V^T$.**

$$V = [V_R \quad V_N], U = [U_C \quad U_{LN}], \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}$$

- **V is an orthogonal matrix, $V^{-1} = V^T$:**

$$A = U\Sigma V^T \rightarrow AV = U\Sigma$$

$$AV = A[V_R \quad V_N] = [AV_R \quad AV_N] = [AV_R \quad 0]$$

$$U = [U_C \quad U_{LN}] \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix} = [U_C\hat{\Sigma} \quad 0]$$

$$[AV_R \quad 0] = [U_C\hat{\Sigma} \quad 0] \rightarrow A\vec{v}_i = \sigma_i\vec{u}_i, \forall i \leq r$$

- **The singular vectors $\vec{u}_i$ are derived directly from the singular vectors $\vec{v}_i$ (for $\sigma_i \neq 0$).**

Relationship between SVD and Four subspaces

$$A = U\Sigma V^T \rightarrow V = [V_R \quad V_N], U = [U_C \quad U_{LN}], \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}$$

- **The $r = rank(A)$ singular values σ_i in $\hat{\Sigma}$ are the square roots of the non-zero eigenvalues of $A^T A$, $\sigma_i = +\sqrt{\lambda_i}$ (repeated for multiplicities over 1).**
- **The r singular vectors $\vec{v}_i$ of V_R are orthonormal eigenvectors of $A^T A$ with non-zero eigenvalues; they are a basis for $\boldsymbol{R}(A)$.**
- **The r singular vectors of U_C are derived from V_R by $\vec{u}_i = (1/\sigma_i)A\vec{v}_i$; they are an orthonormal basis for $\boldsymbol{C}(A)$.**
- **The $n - r$ vectors of V_N are any orthonormal basis for $\boldsymbol{N}(A)$.**
- **The $m - r$ vectors of U_{LN} are any orthonormal basis for $\boldsymbol{N}(A^T)$.**

Relationship between SVD and Four subspaces

$$A = U\Sigma V^T \rightarrow V = [V_R \quad V_N], U = [U_C \quad U_{LN}], \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}$$

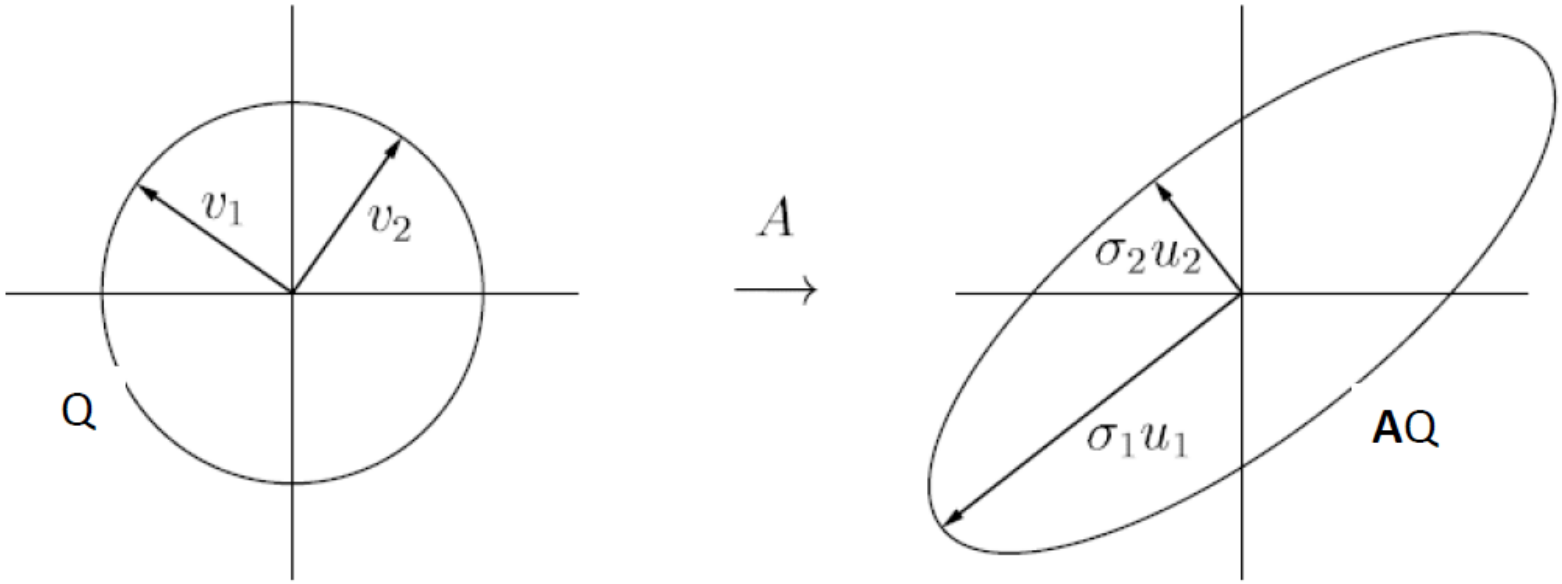
- **The singular value decomposition is *almost* unique.**
- **We could arrange the singular values in $\hat{\Sigma}$ in any order, but by convention we always order them by decreasing magnitude:**

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$$

- **If any of the singular values σ_i are repeated, any orthonormal basis for the eigenspace can be used to chose $\vec{v}_i$; this choice will determine $\vec{u}_i$.**
- **If $nullity(A) > 0$, any orthonormal basis for $N(A)$ can be used for V_N.**
- **If $nullity(A^T) > 0$, any orthonormal basis for $N(A^T)$ can be used for U_{LN}.**

Geometric Interpretation of SVD

- Any *m* x *n* matrix **A** transforms a hypersphere Q of unit radius (called as unit sphere) in $\mathcal{R}^n$ into a hyperellipsoid in $\mathcal{R}^m$ (assume *m* >= *n*).



Why does **A** transform the hypersphere into a *hyper ellipsoid*?

This is because **A** = $\boldsymbol{U\Sigma V^T}$

- V^T transforms the hypersphere into another (rotated/reflected) hypersphere.
- Σ stretches the sphere into a hyper ellipsoid whose semi-axes coincide with the coordinate axes as per **V**.
- **U** rotates/reflects the hyper ellipsoid without affecting its shape.
- As any matrix **A** has an SVD decomposition, it will always transform the hypersphere into a hyper ellipsoid.
- If **A** does not have full rank, then some of the semi-axes of the hyper ellipsoidwill have length 0!

Properties of SVD

1. SVD gives us the **best possible** rank-*r* approximation to *any* matrix (it may or may not be a natural image matrix).
   - The best linear approximation to an ellipse is its longest axis.
   - The best 2D approximation to an ellipsoid in 3D is the ellipse spanned by the longest and second-longest axes.
   - And so on!

2. A square matrix **A** is non-singular (i.e. invertible or full-rank) if and only if all its singular values are non-zero.

3. The ratio σ_1/σ_n tells us how close **A** is to being singular. This ratio is called **condition number** of **A**. The larger the condition number, the closer the matrix is to being singular.

Properties of SVD

4. The rank of a rectangular matrix **A** is equal to the number of non-zero singular values.

5. SVD can be used to find the inverse of the square matrix.

6. Absolute value of the determinant of square matrix A is equal to the product of its singular values.

7. SVD can be used to compute pseudo-inverse of a rectangular matrix:

Examples

Finding the SVD

Ex 1: Find the SVD

- **Find the singular value decomposition of the matrix:**

$$A = \begin{bmatrix} 1 & -3 & 0 \\ 3 & -1 & 0 \end{bmatrix}$$

Find the SVD

- **First, find the eigenvalues of A^TA:**

$$A = \begin{bmatrix} 1 & -3 & 0 \\ 3 & -1 & 0 \end{bmatrix} \rightarrow A^TA = \begin{bmatrix} 1 & 3 \\ -3 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 \\ 3 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 10 & -6 & 0 \\ -6 & 10 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|\lambda I - A^TA| = \begin{vmatrix} \lambda - 10 & 6 & 0 \\ 6 & \lambda - 10 & 0 \\ 0 & 0 & \lambda \end{vmatrix} = \lambda \begin{vmatrix} \lambda - 10 & 6 \\ 6 & \lambda - 10 \end{vmatrix}$$

$$= \lambda\left((\lambda - 10)^2 - 6^2\right) = \lambda(\lambda^2 - 20\lambda + 100 - 36) = \lambda(\lambda - 16)(\lambda - 4)$$

$$|\lambda I - A^TA| = 0 \rightarrow \lambda(\lambda - 16)(\lambda - 4) = 0 \rightarrow \lambda = 16,4,0$$

$$\sigma^2 = \lambda_i \rightarrow \sigma = 4,2 \rightarrow \Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$$

Find the SVD

$$A^T A = \begin{bmatrix} 10 & -6 & 0 \\ -6 & 10 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \lambda = 16,4,0$$

- **Then, find the eigenvectors of $A^T A$:**

$$\lambda = 16: \lambda I - A^T A = \begin{bmatrix} 6 & 6 & 0 \\ 6 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \to \vec{x} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \to \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$\lambda = 4: \lambda I - A^T A = \begin{bmatrix} -6 & 6 & 0 \\ 6 & -6 & 0 \\ 0 & 0 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \to \vec{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \to \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$\lambda = 0: \lambda I - A^T A = \begin{bmatrix} 10 & 6 & 0 \\ 6 & 10 & 0 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \to \vec{x} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Find the SVD

- **Then, find the eigenvectors of $A^T A$:**

$$\lambda_1 = 16 \rightarrow \sigma_1 = 4, \vec{v}_1 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \lambda_2 = 4 \rightarrow \sigma_2 = 2, \vec{v}_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \lambda_3 = 0 \rightarrow \vec{v}_n = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$V = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Find the SVD

- **Finally, compute** AV:

$$AV = \begin{bmatrix} 1 & -3 & 0 \\ 3 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -4/\sqrt{2} & -2/\sqrt{2} & 0 \\ -4/\sqrt{2} & 2/\sqrt{2} & 0 \end{bmatrix}$$

$$AV = U\Sigma \rightarrow \begin{bmatrix} -4/\sqrt{2} & -2/\sqrt{2} & 0 \\ -4/\sqrt{2} & 2/\sqrt{2} & 0 \end{bmatrix} = U \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \rightarrow U = \begin{bmatrix} -1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

Left singular vectors $u_i = \dfrac{Av_i}{\sigma_i}$

Find the SVD

- **Find the singular value decomposition of the matrix:**

$$A = \begin{bmatrix} 1 & -3 & 0 \\ 3 & -1 & 0 \end{bmatrix}$$

$$A = U\Sigma V^T$$

$$U = \begin{bmatrix} -1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}, \Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}, V = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Ex 2:

When is $A = U\Sigma V^{\mathrm{T}}$ (singular values) the *same* as $X\Lambda X^{-1}$ (eigenvalues)?

Solution A needs orthonormal eigenvectors to allow $X = U = V$. A also needs eigenvalues $\lambda \geq 0$ if $\Lambda = \Sigma$. So A must be a ***positive semidefinite*** (***or definite***) ***symmetric matrix***. Only then will $A = X\Lambda X^{-1}$ which is also $Q\Lambda Q^{\mathrm{T}}$ coincide with $A = U\Sigma V^{\mathrm{T}}$.

Ex 3

If $A = \boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}$ (rank 1) with unit vectors $\boldsymbol{x}$ and $\boldsymbol{y}$, what is the SVD of A?

Solution The reduced SVD in (2) is exactly $\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}$, with rank $r = 1$. It has $\boldsymbol{u}_1 = \boldsymbol{x}$ and $\boldsymbol{v}_1 = \boldsymbol{y}$ and $\sigma_1 = 1$. For the full SVD, complete $\boldsymbol{u}_1 = \boldsymbol{x}$ to an orthonormal basis of $\boldsymbol{u}$'s, and complete $\boldsymbol{v}_1 = \boldsymbol{y}$ to an orthonormal basis of $\boldsymbol{v}$'s. No new σ's, only $\sigma_1 = 1$.

Ex 4: An extreme example

- Find the singular value decomposition (matrices U, Σ, V) for the matrix A.

$$A = \begin{bmatrix} 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{2} & 0 \\ 0 & 0 & 0 & \mathbf{3} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Ex 4

$A^{\mathrm{T}}A$ and AA^{T} are diagonal (with easy eigenvectors, but in different orders):

$$A^{\mathrm{T}}A = \begin{bmatrix} \mathbf{0} & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{4} & 0 \\ 0 & 0 & 0 & \mathbf{9} \end{bmatrix} \qquad AA^{\mathrm{T}} = \begin{bmatrix} \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{4} & 0 & 0 \\ 0 & 0 & \mathbf{9} & 0 \\ 0 & 0 & 0 & \mathbf{0} \end{bmatrix}$$

Their eigenvectors ($\boldsymbol{u}$'s for AA^{T} and $\boldsymbol{v}$'s for $A^{\mathrm{T}}A$) go in decreasing order $\sigma_1^2 > \sigma_2^2 > \sigma_3^2$ of the eigenvalues. Those eigenvalues are $\sigma^2 = 9, 4, 1$.

$$U = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & \mathbf{1} & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} \end{bmatrix} \qquad \Sigma = \begin{bmatrix} \mathbf{3} & & & \\ & \mathbf{2} & & \\ & & \mathbf{1} & \\ & & & 0 \end{bmatrix} \qquad V = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & \mathbf{1} & 0 \\ 0 & \mathbf{1} & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Ex 4

eigenvalues $\lambda = 0, 0, 0, 0$ all zero !
only one eigenvector $(1, 0, 0, 0)$
singular values $\sigma = \mathbf{3}, \mathbf{2}, \mathbf{1}$
singular vectors are columns of I

Their eigenvectors ($\boldsymbol{u}$'s for AA^{T} and $\boldsymbol{v}$'s for $A^{\mathrm{T}}A$) go in decreasing order $\sigma_1^2 > \sigma_2^2 > \sigma_3^2$ of the eigenvalues. Those eigenvalues are $\sigma^2 = 9, 4, 1$.

$$U = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & \mathbf{1} & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} \end{bmatrix} \qquad \Sigma = \begin{bmatrix} \mathbf{3} & & & \\ & \mathbf{2} & & \\ & & \mathbf{1} & \\ & & & 0 \end{bmatrix} \qquad V = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & \mathbf{1} & 0 \\ 0 & \mathbf{1} & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Those first columns $\boldsymbol{u}_1$ and $\boldsymbol{v}_1$ have 1's in positions 3 and 4. Then $\boldsymbol{u}_1 \sigma_1 \boldsymbol{v}_1^{\mathrm{T}}$ picks out the biggest number $A_{34} = 3$ in the original matrix A. The three rank-one matrices in the SVD come (for this extreme example) exactly from the numbers $3, 2, 1$ in A.

$$A = U\Sigma V^{\mathrm{T}} = 3\boldsymbol{u}_1\boldsymbol{v}_1^{\mathrm{T}} + 2\boldsymbol{u}_2\boldsymbol{v}_2^{\mathrm{T}} + 1\boldsymbol{u}_3\boldsymbol{v}_3^{\mathrm{T}}.$$

Ex 5: Singular value stability vs eigenvalue instability

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ \frac{1}{60,000} & 0 & 0 & 0 \end{bmatrix}$$

That change by only $1/60,000$ produces a much bigger jump in the eigenvalues of A

$$\lambda = 0, 0, 0, 0 \text{ to } \boldsymbol{\lambda} = \frac{1}{10}, \frac{i}{10}, \frac{-1}{10}, \frac{-i}{10}$$

The four eigenvalues moved from zero onto a circle around zero. The circle has radius $\frac{1}{10}$ when the new entry is only $1/60,000$. This shows serious instability of eigenvalues when AA^{T} is far from $A^{\mathrm{T}}A$. At the other extreme, if $A^{\mathrm{T}}A = AA^{\mathrm{T}}$ (a "normal matrix") the eigenvectors of A are orthogonal and the eigenvalues of A are totally stable.

By contrast, **the singular values of any matrix are stable**. They don't change more than the change in A. In this example, the new singular values are $\mathbf{3}, \mathbf{2}, \mathbf{1}$, and $\mathbf{1/60,000}$. The matrices U and V stay the same. The new fourth piece of A is $\sigma_4 \boldsymbol{u}_4 \boldsymbol{v}_4^{\mathrm{T}}$, with fifteen zeros and that small entry $\sigma_4 = 1/60,000$.

Section 7.3 & 7.4

Principal Component Analysis & Geometry of the SVD

Reduced-rank approximations.

- **Most data and image matrices are either full row rank or full column rank; the rank for matrices with both many rows and many columns can be large:**

$$A \in \mathbb{R}^{m\times n}, rank(A) = \min(m, n)$$

- **However, many data and image matrices can be well approximated by a matrix of significantly lower rank:**

$$\hat{A}_k \in \mathbb{R}^{m\times n}, rank(\hat{A}_k) = k \ll m, n, \hat{A}_k \approx A$$

- **This low-rank approximation may be thought of a "big picture" view of the matrix, capturing the essential features, but missing subtle details and/or noise.**

Reduced-rank approximations.

- **We begin with the view of as a sum of rank-1 matrices, expressed in terms of the singular values/vectors:**

$$A = \sum_{i=1}^{r} \sigma_i \vec{u}_i \vec{v}_i^T \, , \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$$

- **$\{\vec{u}_i\}$ and $\{\vec{v}_i\}$ are orthonormal sets, so all $\vec{u}_i \vec{v}_i^T$ will be of similar magnitudes; the contribution of each term to the overall sum will be dominated by σ_i. We just truncate the sum:**

$$\hat{A}_k \equiv \sum_{i=1}^{k} \sigma_i \vec{u}_i \vec{v}_i^T$$

Reduced-rank approximations.

- In summation notation we have: $\hat{A}_k \equiv \sum_{i=1}^{k} \sigma_i \vec{u}_i \vec{v}_i^T$
- The full SVD was represented as:

$$A = U\Sigma V^T, V = [V_R \quad V_N], U = [U_C \quad U_{LN}], \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

- We can define a reduced-rank SVD simply by changing Σ; we zero out all the singular values after the first k:

$$\hat{A}_k = U\Sigma_k V^T, \Sigma_k = \begin{bmatrix} \hat{\Sigma}_k & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma}_k = \begin{bmatrix} \sigma_1' & & \\ & \ddots & \\ & & \sigma_k' \end{bmatrix}, \sigma_i' = \sigma_i, \forall i \leq k, \sigma_i' = 0, \forall i > k$$

Reduced-rank approximations.

- In a simplified notation we can write:

$$\hat{A}_k = U\Sigma_k V^T$$

$$\Sigma_k = \begin{bmatrix} \hat{\Sigma}_k & 0 \\ 0 & 0 \end{bmatrix}, \qquad \hat{\Sigma}_k = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_k \end{bmatrix}$$

- $\hat{A}_k$ is the **best** rank-k approximation of A, in a "least-squares" sense:

$$\hat{A}_k = \min_{X, rank(X)=k} \left(\sum_{i,j} \left(A_{ij} - X_{ij}\right)^2 \right)$$

Efficient storage of A and $\hat{A}_k$

- **The full SVD was represented as:**

$$A = U\Sigma V^T, V = [V_R \quad V_N], U = [U_C \quad U_{LN}]\, \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

  - **Multiplying out the matrices, we find that both null spaces can be ignored; only the row and column space vectors are needed to perfectly reconstruct A:**

$$A = [U_C \quad U_{LN}] \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_R^T \\ V_N^T \end{bmatrix} = [U_C \quad U_{LN}] \begin{bmatrix} \hat{\Sigma} V_R^T \\ 0 \end{bmatrix} = U_C \hat{\Sigma} V_R^T$$

- **A rank-r matrix can be "stored" as r vectors of length m, r vectors of length n and r singular values: total storage is** $mr + nr + r = r(m + n + 1)$

$$r \ll \min(m, n) \rightarrow r(m + n + 1) \ll mn$$

$$r = \min(m, n) \rightarrow r(m + n + 1) > mn$$

Efficient storage of A and $\hat{A}_k$

- The reduced-rank SVD can be represented as:

$$\hat{A}_k = U\Sigma_k V^T, V = [\hat{V}_k \quad \hat{V}_N], U = [\hat{U}_k \quad \hat{U}_{LN}], \Sigma_k = \begin{bmatrix} \hat{\Sigma}_k & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma}_k = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_k \end{bmatrix}$$

- Here $\hat{V}_k$ and $\hat{U}_k$ contain only the first columns of V_R and U_C; the others have been added to V_N and U_{LN} to form $\hat{V}_N$ and $\hat{U}_{LN}$. Then:

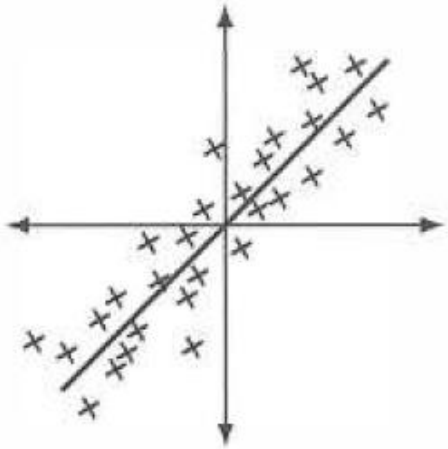
$$\hat{A}_k = [\hat{U}_k \quad \hat{U}_{LN}] \begin{bmatrix} \hat{\Sigma}_k & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{V}_k^T \\ \hat{V}_N^T \end{bmatrix} = [\hat{U}_k \quad \hat{U}_{LN}] \begin{bmatrix} \hat{\Sigma}_k \hat{V}_k^T \\ \hat{V}_N^T \end{bmatrix} = \hat{U}_k \hat{\Sigma}_k \hat{V}_k^T$$

- $\hat{A}_k$ can be stored as k vectors of length m, k vectors of length n and k singular values: total storage is $mk + nk + k = k(m + n + 1)$:

$$k \ll \min(m, n) \rightarrow k(m + n + 1) \ll mn$$

7.3: Why do we need Principal Component Analysis?

Let me start with a picture instead of numbers. For $m = 2$ variables like age and height, the n points lie in the plane $\mathbf{R}^2$. Subtract the average age and height to center the data. If the n recentered points cluster along a line, *how will linear algebra find that line* ?



A is $\mathbf{2} \times \boldsymbol{n}$ (large nullspace)

AA^{T} is $\mathbf{2} \times \mathbf{2}$ (small matrix)

$A^{\mathrm{T}}A$ is $\boldsymbol{n} \times \boldsymbol{n}$ (large matrix)

Two singular values $\boldsymbol{\sigma_1} > \boldsymbol{\sigma_2} > \mathbf{0}$

Figure 7.2: Data points in A are often close to a line in $\mathbf{R}^2$ or a subspace in $\mathbf{R}^m$.

Let me go more carefully in constructing the data matrix. Start with the measurements in A_0 : the sample data. Find the average (the *mean*) $\mu_1, \mu_2, \ldots, \mu_m$ of each row. *Subtract each mean μ_i from row i to center the data.* The average along each row is now zero, for the centered matrix A. So the point $(0,0)$ in Figure 7.2 is now the true center of the n points.

The "sample covariance matrix" is defined by $\boldsymbol{S = \dfrac{AA^{\mathrm{T}}}{n-1}}$.

A shows the distance $a_{ij} - \mu_i$ from each measurement to the row average μ_i.

$(AA^{\mathrm{T}})_{11}$ and $(AA^{\mathrm{T}})_{22}$ show the **sum of squared distances (sample variances s_1^2, s_2^2)**.

$(AA^{\mathrm{T}})_{12}$ shows the **sample covariance s_{12}** $=$ (row 1 of A)$\cdot$(row 2 of A).

The variance is a key number throughout statistics. An average exam score $\mu = 85$ tells you it was a decent exam. A variance of $s^2 = 25$ (standard deviation $s = 5$) means that most grades were in the 80's: closely packed. A sample variance $s^2 = 225$ ($s = 15$) means that grades were widely scattered. Chapter 12 explains variances.

Example: Six math and history scores (0 means in each row)

Example 1 **Six math and history scores (notice the zero mean in each row)**

$$A = \begin{bmatrix} 3 & -4 & 7 & 1 & -4 & -3 \\ 7 & -6 & 8 & -1 & -1 & -7 \end{bmatrix} \text{ has sample covariance } S = \frac{AA^{\mathrm{T}}}{5} = \begin{bmatrix} 20 & 25 \\ 25 & 40 \end{bmatrix}.$$

- The two rows of A are highly correlated because S12 =25 which means above average math went with above average history.

The eigenvalues of S are near 57 and 3. So the first rank one piece $\sqrt{57}\,\boldsymbol{u}_1\boldsymbol{v}_1^{\mathrm{T}}$ is much larger than the second piece $\sqrt{3}\,\boldsymbol{u}_2\boldsymbol{v}_2^{\mathrm{T}}$. **The leading eigenvector $\boldsymbol{u}_1$ shows the direction that you see in the scatter graph of Figure 7.2.** That eigenvector is close to $\boldsymbol{u}_1 = (.6, .8)$ and the direction in the graph nearly gives a $6-8-10$ or $3-4-5$ right triangle.

The SVD of A (centered data) shows the dominant direction in the scatter plot.

The second singular vector $\boldsymbol{u}_2$ is perpendicular to $\boldsymbol{u}_1$. The second singular value $\sigma_2 \approx \sqrt{3}$ measures the spread across the dominant line. If the data points in A fell exactly on a line ($\boldsymbol{u}_1$ direction), then σ_2 would be zero. Actually there would only be σ_1.

7.3: Principal component analysis.

- **For a data matrix with each column corresponding to individual "samples" and each row corresponding to a different measurable variable.**
- **If the rows of the matrix are mean-centered (we subtract the average of each row from every element of that row), then the covariance matrix is:**

$$Cov(A) = S = \left(\frac{1}{n-1}\right) AA^T$$

- **Each element of the covariance matrix describes how much a pair of rows change in a correlated manner: positive values mean that the two rows tend to have high (or low) values in the same column; negative values mean that high values in a column for one row usually match with low values for the other.**

7.3: Principal component analysis.

- **For a (row) mean-shifted data matrix, the covariance matrix is:**

$$Cov(A) = S = \left(\frac{1}{n-1}\right) AA^T$$

- **The total variance in the data is the sum of the eigenvalues of the covariance matrix (T= trace of A (diagonal sum)):**

$$T = \lambda_1 + \lambda_2 + \cdots + \lambda_n = \lambda_1 + \lambda_2 + \cdots + \lambda_r, rank\,(A) = r$$

- **The covariance matrix involves the same product used to derive singular values of A:**

$$\lambda_i = \sigma_i^2 \longrightarrow T = (\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2)$$

Principal component analysis.

- **For a (row) mean-shifted data matrix, total variance in the data is given by the sum of the singular values (scaled by the sample size):**

$$\mathrm{T} = (\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2)$$

- **The largest singular values explain most of the variance in the data.**

$$T \approx T_k = \left(\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_k^2\right), k < r$$

- **Just like a low-rank approximation:** $\hat{A}_k = \hat{U}_k \hat{\Sigma}_k \hat{V}_k^T$
- **The singular vectors $\vec{u}_i$ tell us the directions in which the majority of the variance occurs, these are the so-called principal components.**

Principal component analysis.

The singular vectors $\vec{u}_i$ tell us the directions in which the majority of the variance occurs, these are the so-called principal components.

- ✓ **To find a best fit line (through the origin), we follow the single direction of greatest variance: the first principal component.**
- ✓ **To find a best fit plane (through the origin), we consider the two directions of greatest variance: the first and second principal components.**
- ✓ **The first "k" principal components is the k-dimensional subspace which best (in a least-squares sense) satisfies the data.**

Alternative way:

SVD gives us the **best possible** rank-*r* approximation to *any* matrix (it may or may not be a natural image matrix).

- The best linear approximation to an ellipse is its longest axis.
- The best 2D approximation to an ellipsoid in 3D is the ellipse spanned by the longest and second-longest axes.
- And so on!

7.4: Pseudoinverses and the SVD

- **The full SVD was represented as:**

$$A = U\Sigma V^T, V = [V_R \quad V_N], U = [U_C \quad U_{LN}], \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

- **If A is square and nonsingular:**

$$A^{-1} = (U\Sigma V^T)^{-1} = (V^T)^{-1}\Sigma^{-1}U^{-1} = V\Sigma^{-1}U^T$$

$$\Sigma = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{bmatrix} \rightarrow \Sigma^{-1} = \begin{bmatrix} 1/\sigma_1 & & \\ & \ddots & \\ & & 1/\sigma_r \end{bmatrix}$$

Pseudoinverses and the SVD

- **The full SVD was represented as:**

$$A = U\Sigma V^T, V = [V_R \quad V_N], U = [U_C \quad U_{LN}], \Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

- **If A is either non-square or singular, define:**

$$\Sigma^+ \equiv \begin{bmatrix} \hat{\Sigma}^{-1} & 0 \\ 0 & 0 \end{bmatrix}, \hat{\Sigma}^{-1} = \begin{bmatrix} 1/\sigma_1 & & \\ & \ddots & \\ & & 1/\sigma_r \end{bmatrix}$$

$$A^+ \equiv V\Sigma^+U^T = V_R\hat{\Sigma}^{-1}U_C^T$$

Pseudoinverses and the SVD

- **The product AA^+ almost gives the identity:**

$$AA^+ = U\Sigma V^T V\Sigma^+ U^T = U\Sigma\Sigma^+ U^T$$

$$\Sigma\Sigma^+ = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{\Sigma}^{-1} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}, UU^T = I_m$$

$$AA^+ = \begin{bmatrix} U_C & U_{LN} \end{bmatrix} \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U_C^T \\ U_{LN}^T \end{bmatrix} = U_C U_C^T$$

- $P_C = U_C U_C^T = AA^+$ **is the projection matrix onto the column space.**
- **This will be exactly the $(m \times m)$ identity matrix if (and only if) A is full row rank (right inverse).**

Pseudoinverses and the SVD

- **The product A^+A also almost gives the identity:**

$$A^+A = V\Sigma^+U^TU\Sigma V^T = V\Sigma^+\Sigma V^T$$

$$\Sigma^+\Sigma = \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \hat{\Sigma}^{-1} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}, VV^T = I_n$$

$$A^+A = \begin{bmatrix} V_R & V_N \end{bmatrix}\begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} V_R^T \\ V_N^T \end{bmatrix} = V_RV_R^T$$

- $P_R = V_RV_R^T = A^+A$ **is the projection matrix onto the row space.**
- **This will be exactly the $(n \times n)$ identity matrix if (and only if) A is full column rank (left inverse).**

Pseudoinverses and the SVD

- $P_C = U_C U_C^T = AA^+$ is the projection matrix onto the column space.
- The best least-squares solution to an inconsistent system is found by solving:

$$A\vec{x} = proj_{C(A)}\vec{b} = P_C\vec{b} \rightarrow A\vec{x} = AA^+\vec{b}, A^+ \equiv V\Sigma^+U^T$$
$$A\vec{x} = AA^+\vec{b} \rightarrow U_C\hat{\Sigma}V_R^T\vec{x} = U_CU_C^T\vec{b} \rightarrow V_R^T\vec{x} = \hat{\Sigma}^{-1}U_C^T\vec{b}$$
$$V_RV_R^T\vec{x} = V_R\hat{\Sigma}^{-1}U_C^T\vec{b} \rightarrow P_R\vec{x} = V_R\hat{\Sigma}^{-1}U_C^T\vec{b}$$
$$\vec{x}_r = P_R\vec{x} = A^+\vec{b}$$

- $\vec{x}_r$ is the projection of the solution(s) onto the row space of A.
- This is the solution to $A\vec{x} = proj_{C(A)}\vec{b}$ that is in the row space of A.

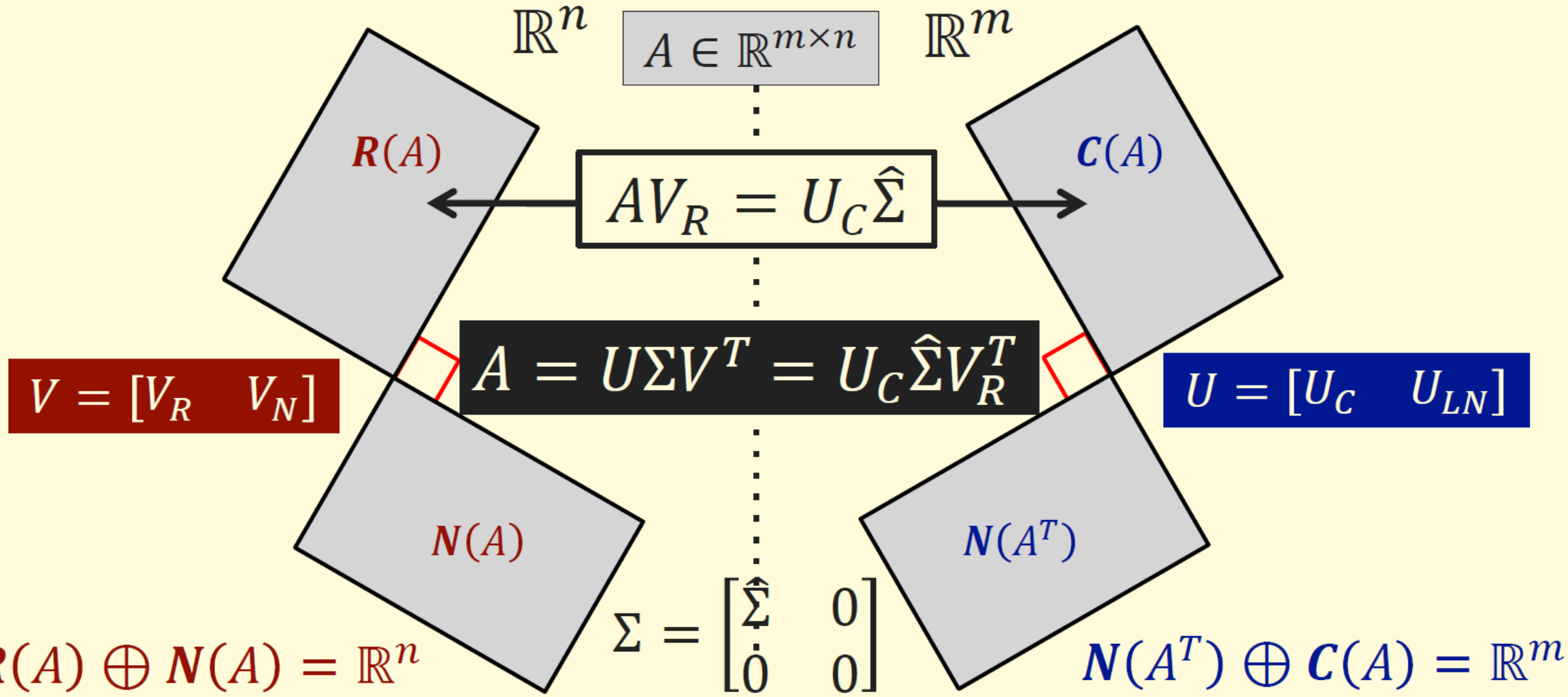
Pseudoinverses and the SVD

- The pseudoinverse gives the solution to $A\vec{x} = proj_{C(A)}\vec{b}$ that is in the row space of A.

$$\vec{x}_r = A^+\vec{b}, A^+ \equiv V\Sigma^+U^T$$

- The general solution is $\vec{x} = \vec{x}_r + \vec{x}_n, \forall \vec{x}_n \in \boldsymbol{N}(A)$.
- The row space and null space are orthogonal: $\vec{x}_r \perp \vec{x}_n$.
- $\vec{x}_r$ is the shortest of any solution: $\|\vec{x}\|^2 = \|\vec{x}_r\|^2 + \|\vec{x}_n\|^2$.
- Given $A\vec{x} = \vec{b}$ the pseudoinverse gives the shortest possible solution to the system $A\vec{x} = \vec{p}$ that has a right-hand side $\vec{p}$ that is the shortest possible distance from $\vec{b}$.

The SVD and the big picture.



Examples

Using the SVD

Ex 1: Find a low-rank approximation

- Find the best (in a least-squares sense) rank-2
- Approximation to the following matrix with a given SVD:

$$\begin{bmatrix} 3 & -2.25 & -1.75 & 3 \\ 3 & -1.75 & -2.25 & 3 \\ 2.05 & -3 & -3 & 1.95 \\ 1.95 & -3 & -3 & 2.05 \end{bmatrix}$$

$$A = \begin{bmatrix} 1/2 & 1/2 & -1/\sqrt{2} & 0 \\ 1/2 & 1/2 & 1/\sqrt{2} & 0 \\ 1/2 & -1/2 & 0 & -1/\sqrt{2} \\ 1/2 & -1/2 & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/10 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 & 1/2 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 0 & 0 & 1/\sqrt{2} \end{bmatrix}$$

Ex 1: Find a low-rank approximation

- **Zero out all but the two largest singular values**

$$\hat{A}_2 = \begin{bmatrix} 1/2 & 1/2 & -1/\sqrt{2} & 0 \\ 1/2 & 1/2 & 1/\sqrt{2} & 0 \\ 1/2 & -1/2 & 0 & -1/\sqrt{2} \\ 1/2 & -1/2 & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 & 1/2 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 0 & 0 & 1/\sqrt{2} \end{bmatrix}$$

Find a low-rank approximation

- **Zero out all but the two largest singular values**

$$\hat{A}_2 = \begin{bmatrix} 1/2 & 1/2 & -1/\sqrt{2} & 0 \\ 1/2 & 1/2 & 1/\sqrt{2} & 0 \\ 1/2 & -1/2 & 0 & -1/\sqrt{2} \\ 1/2 & -1/2 & 0 & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 5 & -5 & -5 & 5 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & -2 & -2 & 3 \\ 3 & -2 & -2 & 3 \\ 2 & -3 & -3 & 2 \\ 2 & -3 & -3 & 2 \end{bmatrix}$$

Find a low-rank approximation

- **Find the best (in a least-squares sense) rank-2 approximation to the following matrix with a given SVD:**

$$\hat{A}_2 = \begin{bmatrix} 3 & -2 & -2 & 3 \\ 3 & -2 & -2 & 3 \\ 2 & -3 & -3 & 2 \\ 2 & -3 & -3 & 2 \end{bmatrix}, \qquad A = \begin{bmatrix} 3 & -2.25 & -1.75 & 3 \\ 3 & -1.75 & -2.25 & 3 \\ 2.05 & -3 & -3 & 1.95 \\ 1.95 & -3 & -3 & 2.05 \end{bmatrix}$$

$$\Delta = (A - \hat{A}_2) = \begin{bmatrix} 0 & -0.25 & +0.25 & 3 \\ 0 & +0.25 & -0.25 & 3 \\ +0.05 & 0 & 0 & -0.05 \\ -0.05 & 0 & 0 & +0.05 \end{bmatrix}$$

Ex 2: Solve with a pseudoinverse

- **Find the general least-squares solution to the following system of equations, given the SVD of the coefficient matrix:**

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix}$$

$$A = U\Sigma V^T, U = \begin{bmatrix} -1/2 & 1/2 & 1/\sqrt{2} & 0 \\ -1/2 & 1/2 & -1/\sqrt{2} & 0 \\ -1/2 & -1/2 & 0 & -1/\sqrt{2} \\ -1/2 & -1/2 & 0 & 1/\sqrt{2} \end{bmatrix}, \Sigma = \begin{bmatrix} 4\sqrt{3} & 0 & 0 \\ 0 & 2\sqrt{2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, V = \begin{bmatrix} -1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ -1/\sqrt{3} & 0 & -2/\sqrt{6} \\ -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}$$

Solve with a pseudoinverse

- **The least-squares solution from the row space is given by the pseudoinverse:**

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \rightarrow \vec{x}_r = A^+\vec{b}, A^+ = V\Sigma^+U^T$$

$$A = U\Sigma V^T, U = \begin{bmatrix} -1/2 & 1/2 & 1/\sqrt{2} & 0 \\ -1/2 & 1/2 & -1/\sqrt{2} & 0 \\ -1/2 & -1/2 & 0 & -1/\sqrt{2} \\ -1/2 & -1/2 & 0 & 1/\sqrt{2} \end{bmatrix}, \Sigma = \begin{bmatrix} 4\sqrt{3} & 0 & 0 \\ 0 & 2\sqrt{2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, V = \begin{bmatrix} -1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ -1/\sqrt{3} & 0 & -2/\sqrt{6} \\ -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}$$

Solve with a pseudoinverse

- **The least-squares solution from the row space is given by the pseudoinverse:**

$$A^+ = V\Sigma^+U^T = \begin{bmatrix} -1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ -1/\sqrt{3} & 0 & -2/\sqrt{6} \\ -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 1/4\sqrt{3} & 0 & 0 & 0 \\ 0 & 1/2\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1/2 & -1/2 & -1/2 & -1/2 \\ 1/2 & 1/2 & -1/2 & -1/2 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 0 & 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} -1/12 & -1/4 & 0 & 0 \\ -1/12 & 0 & 0 & 0 \\ -1/12 & 1/4 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1/2 & -1/2 & -1/2 & -1/2 \\ 1/2 & 1/2 & -1/2 & -1/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -1/12 & -1/12 & 1/6 & 1/6 \\ 1/24 & 1/24 & 1/24 & 1/24 \\ 1/6 & 1/6 & -1/12 & -1/12 \end{bmatrix}$$

Solve with a pseudoinverse

- **The least-squares solution from the row space is given by the pseudoinverse:**

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \rightarrow \vec{x}_r = A^+\vec{b}, A^+ = \begin{bmatrix} -1/12 & -1/12 & 1/6 & 1/6 \\ 1/24 & 1/24 & 1/24 & 1/24 \\ 1/6 & 1/6 & -1/12 & -1/12 \end{bmatrix}$$

$$\vec{x}_r = A^+\vec{b}, A^+ = \begin{bmatrix} -1/12 & -1/12 & 1/6 & 1/6 \\ 1/24 & 1/24 & 1/24 & 1/24 \\ 1/6 & 1/6 & -1/12 & -1/12 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/12 \\ 5/12 \\ 11/12 \end{bmatrix}$$

Solve with a pseudoinverse

- **The general least-squares solution is given by adding the null space.**

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \rightarrow \vec{x} = \vec{x}_r + s\vec{x}_n, \forall s, \vec{x}_r = A^+\vec{b}, A\vec{x}_n = \vec{0}$$

$$A = U\Sigma V^T, V = \begin{bmatrix} -1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ -1/\sqrt{3} & 0 & -2/\sqrt{6} \\ -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix} \rightarrow \vec{x}_n = \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}$$

Solve with a pseudoinverse

- **The least-squares solution from the row space is given by the pseudoinverse:**

$$A\vec{x} = \vec{b}, A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \rightarrow \vec{x}_r = \begin{bmatrix} -1/12 \\ 5/12 \\ 11/12 \end{bmatrix}, \vec{x}_n = \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}$$

$$\vec{x} = \vec{x}_r + s\vec{x}_n = \begin{bmatrix} -1/12 \\ 5/12 \\ 11/12 \end{bmatrix} + s \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \forall s \in \mathbb{R}$$